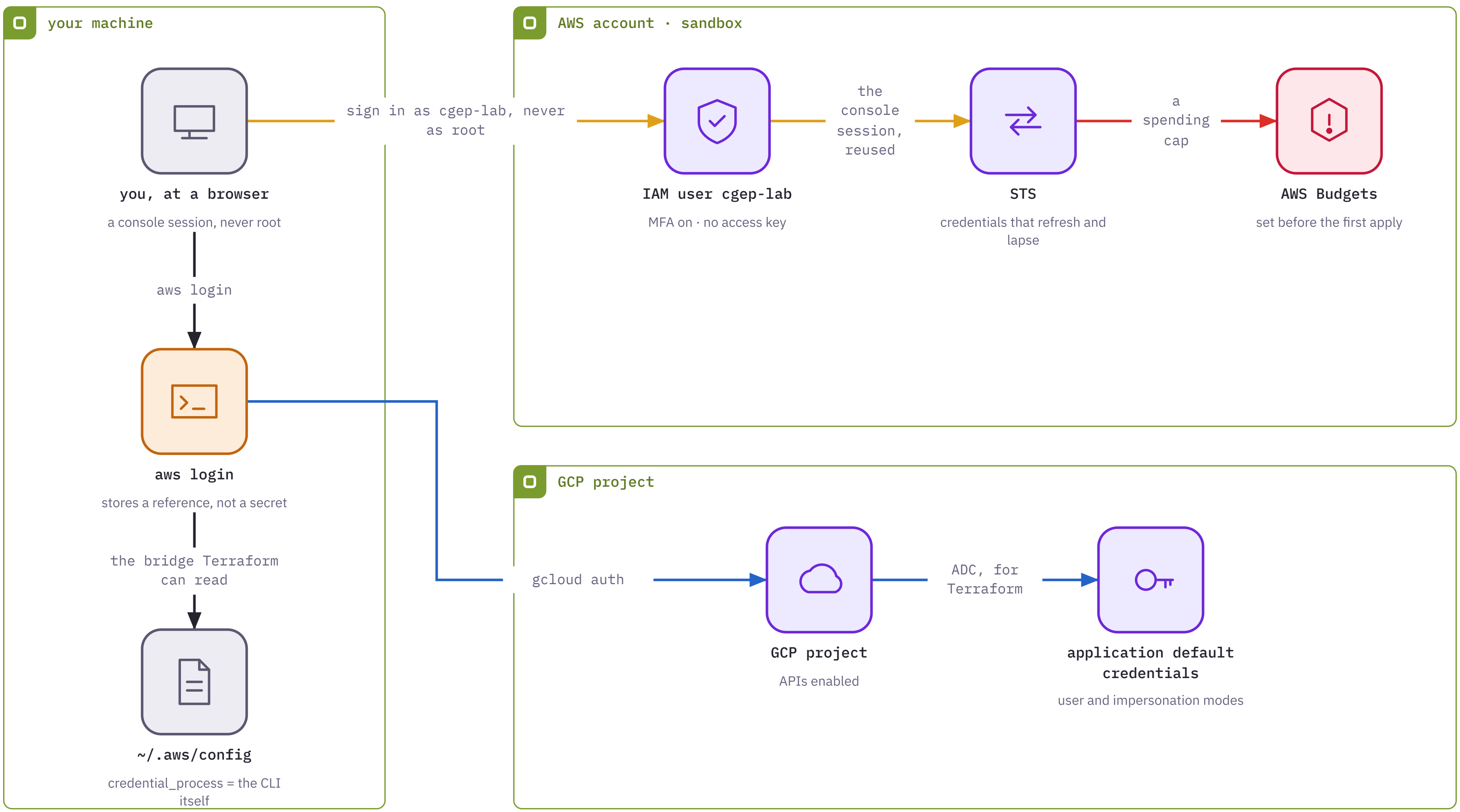
CGE-P LAB CURRICULUM · V2

Lab 0.1 — proving who you are, briefly

AWS identity · GCP identity · on your disk · the spending cap

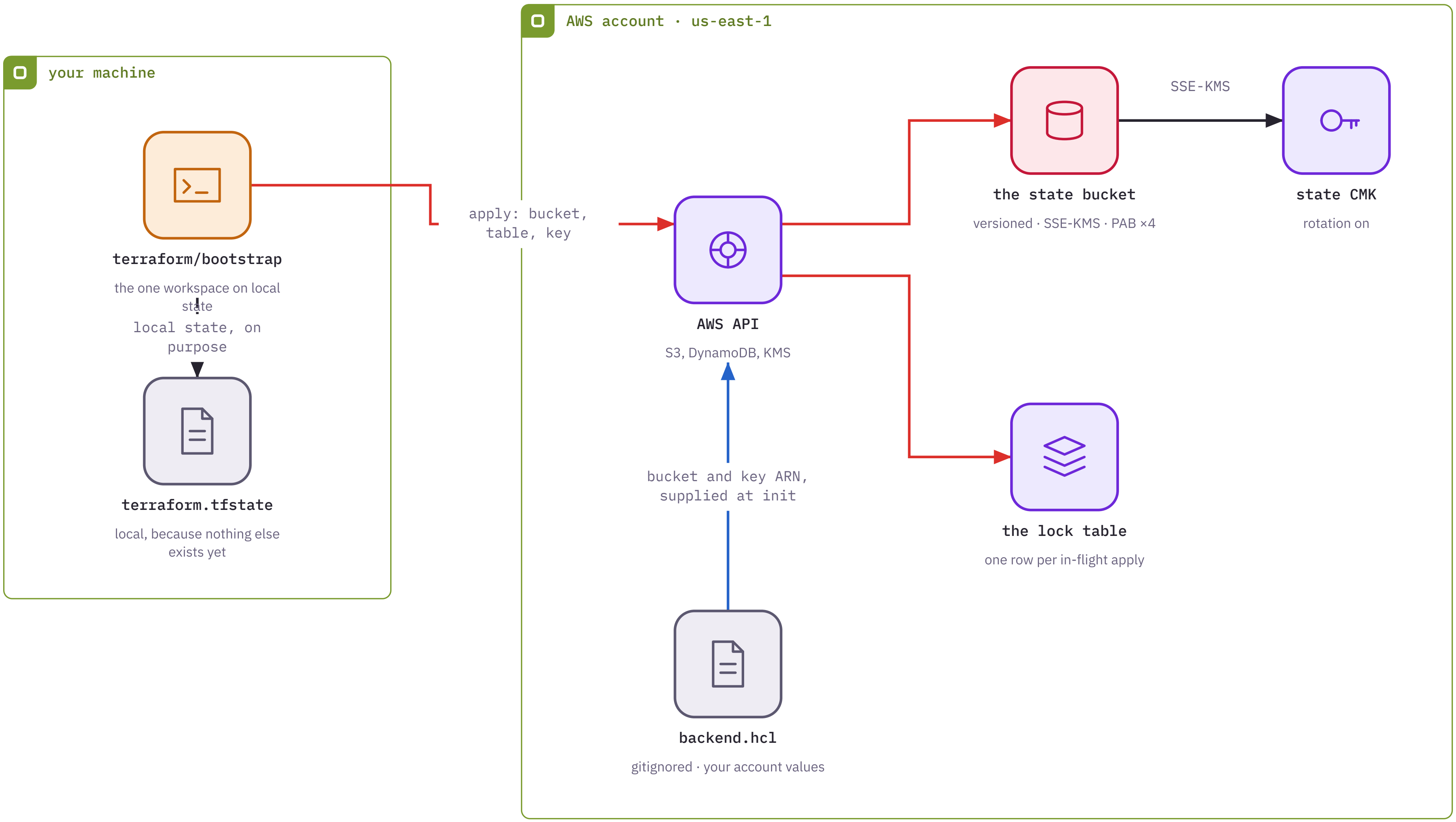
Nothing is deployed. You are establishing an identity for both clouds that expires on its own, and putting a ceiling on what everything after this can cost you.



Lab 2.2 — where state goes to live

provisioning · partial backend config · encryption and local state

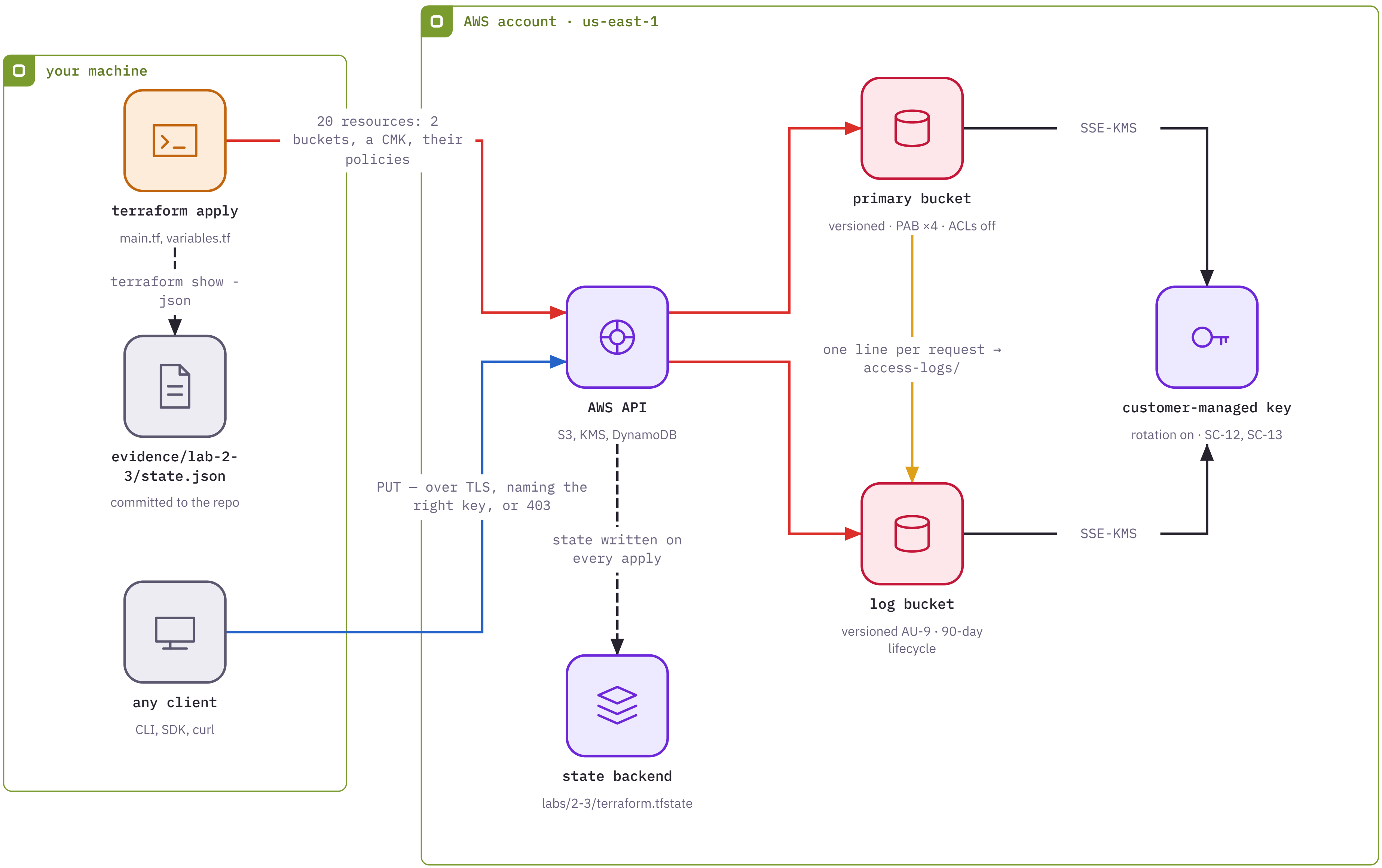
Terraform's record of what it built stops living on your laptop. One workspace still keeps local state, deliberately: it is the thing that creates the place state goes.



Lab 2.3 — the compliant S3 primitive

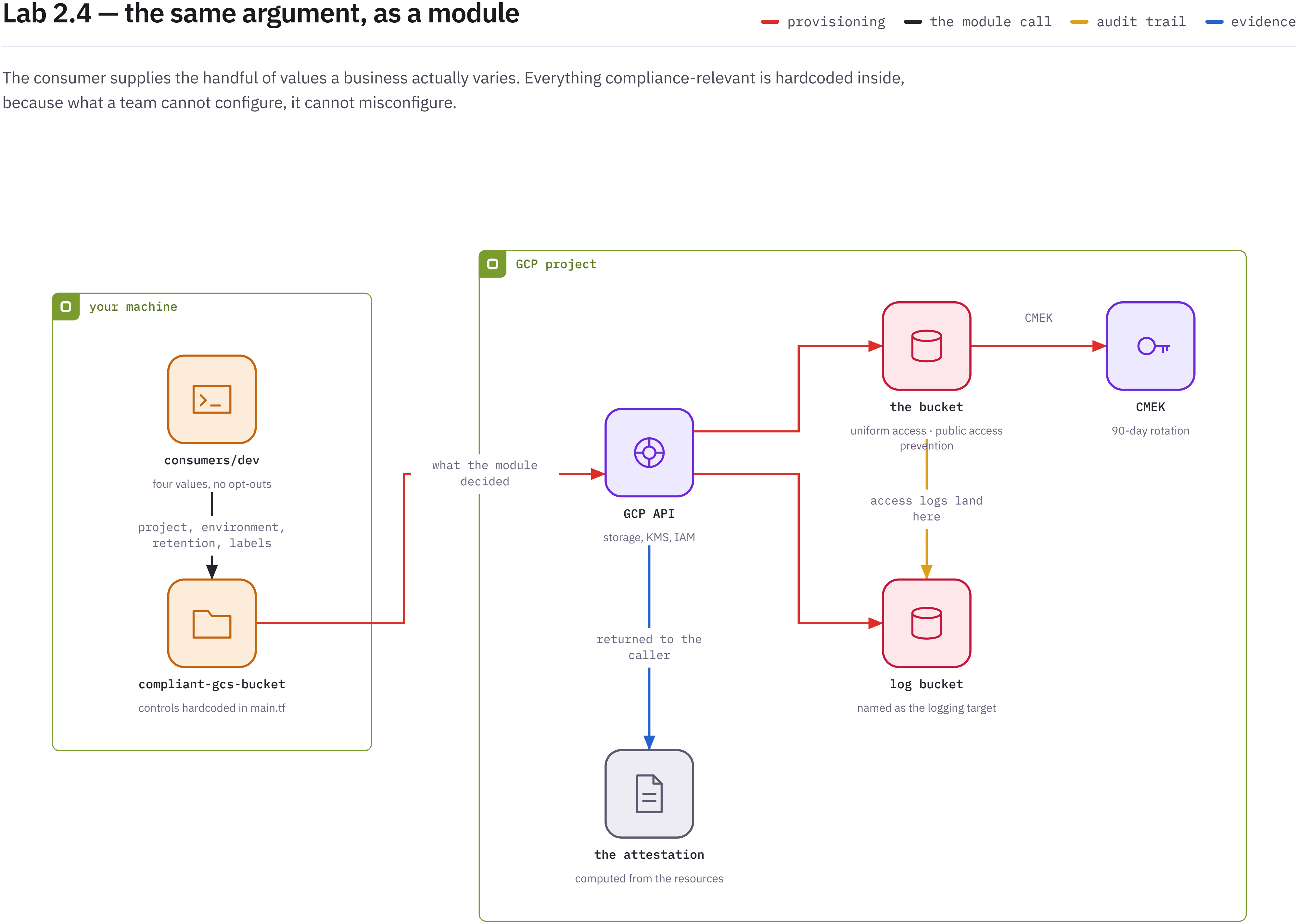
provisioning · request path · audit trail · key use and state

You write Terraform once. AWS then enforces the controls on every request that follows: a request without TLS is refused by the bucket itself, and an upload naming the wrong key is refused before an object exists.



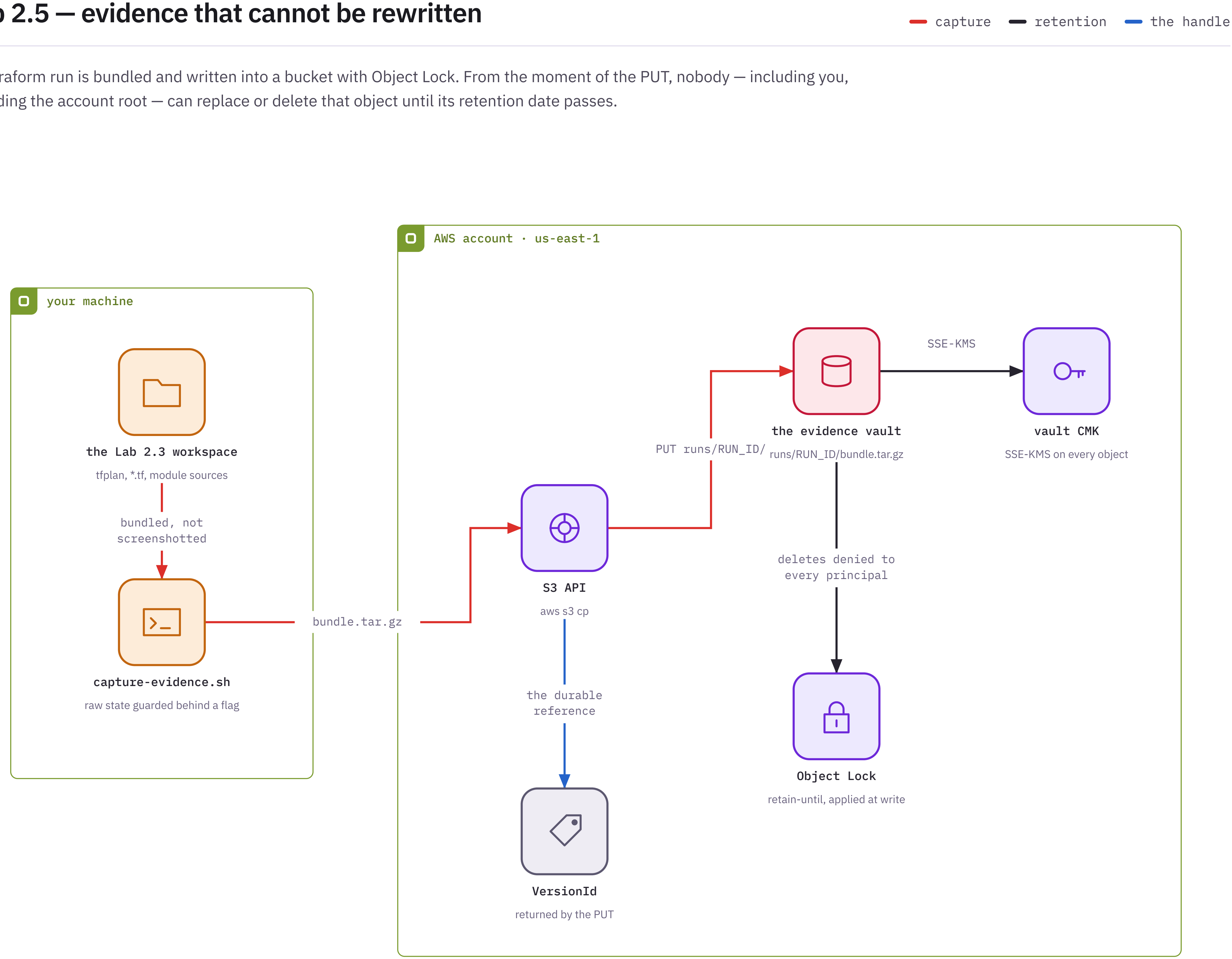
Lab 2.4 — the same argument, as a module

The consumer supplies the handful of values a business actually varies. Everything compliance-relevant is hardcoded inside, because what a team cannot configure, it cannot misconfigure.



Lab 2.5 — evidence that cannot be rewritten

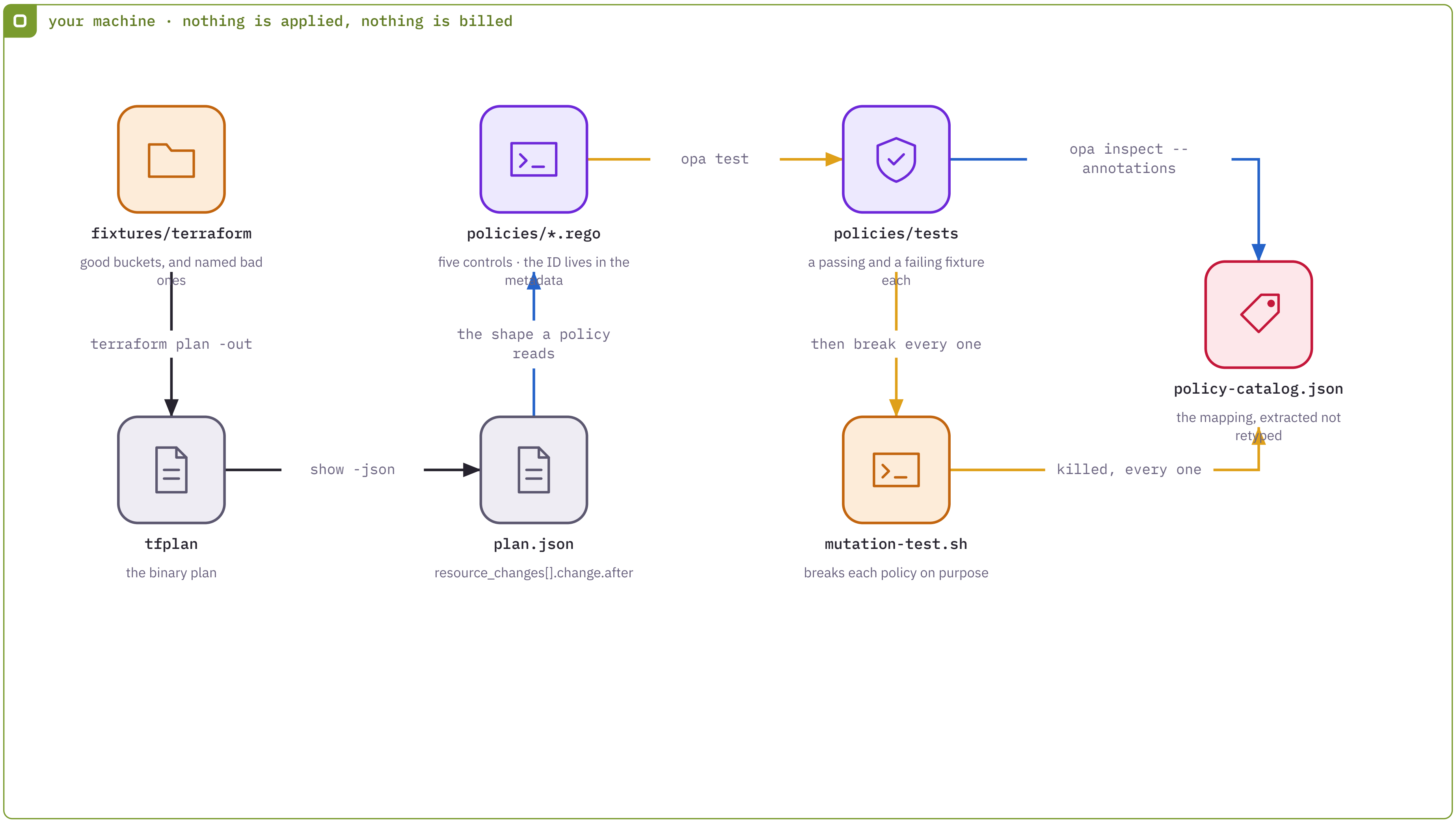
A Terraform run is bundled and written into a bucket with Object Lock. From the moment of the PUT, nobody — including you, including the account root — can replace or delete that object until its retention date passes.



Lab 3.3 — checking before anything exists

plan-time · the policies · testing the tests

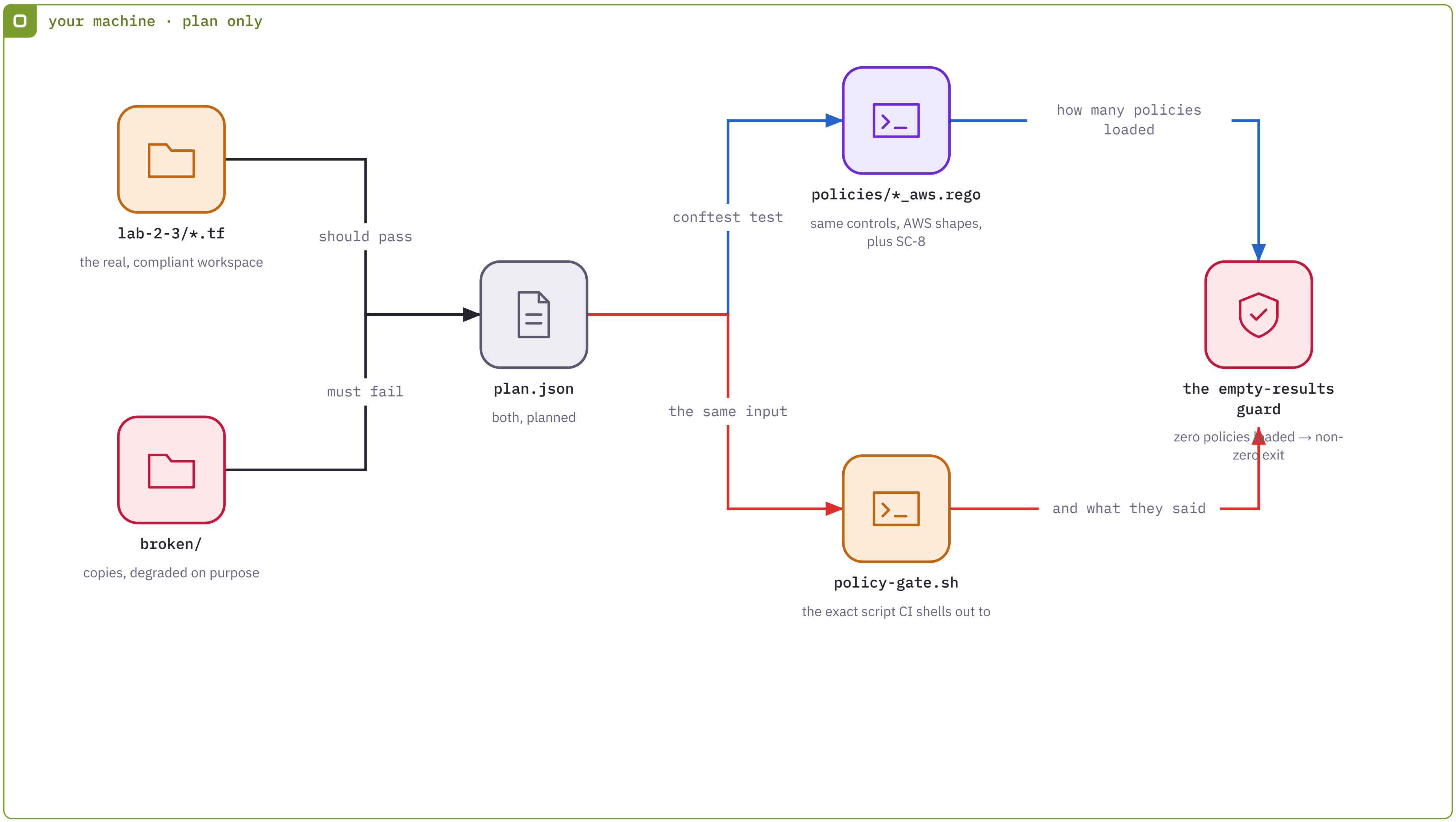
Nothing is applied. Terraform produces a plan, the plan becomes JSON, and the policies read that JSON — so a bad bucket is caught before it is a bucket.



Lab 3.4 — the gate, and proof it can fail

plan-time · the check · the gate

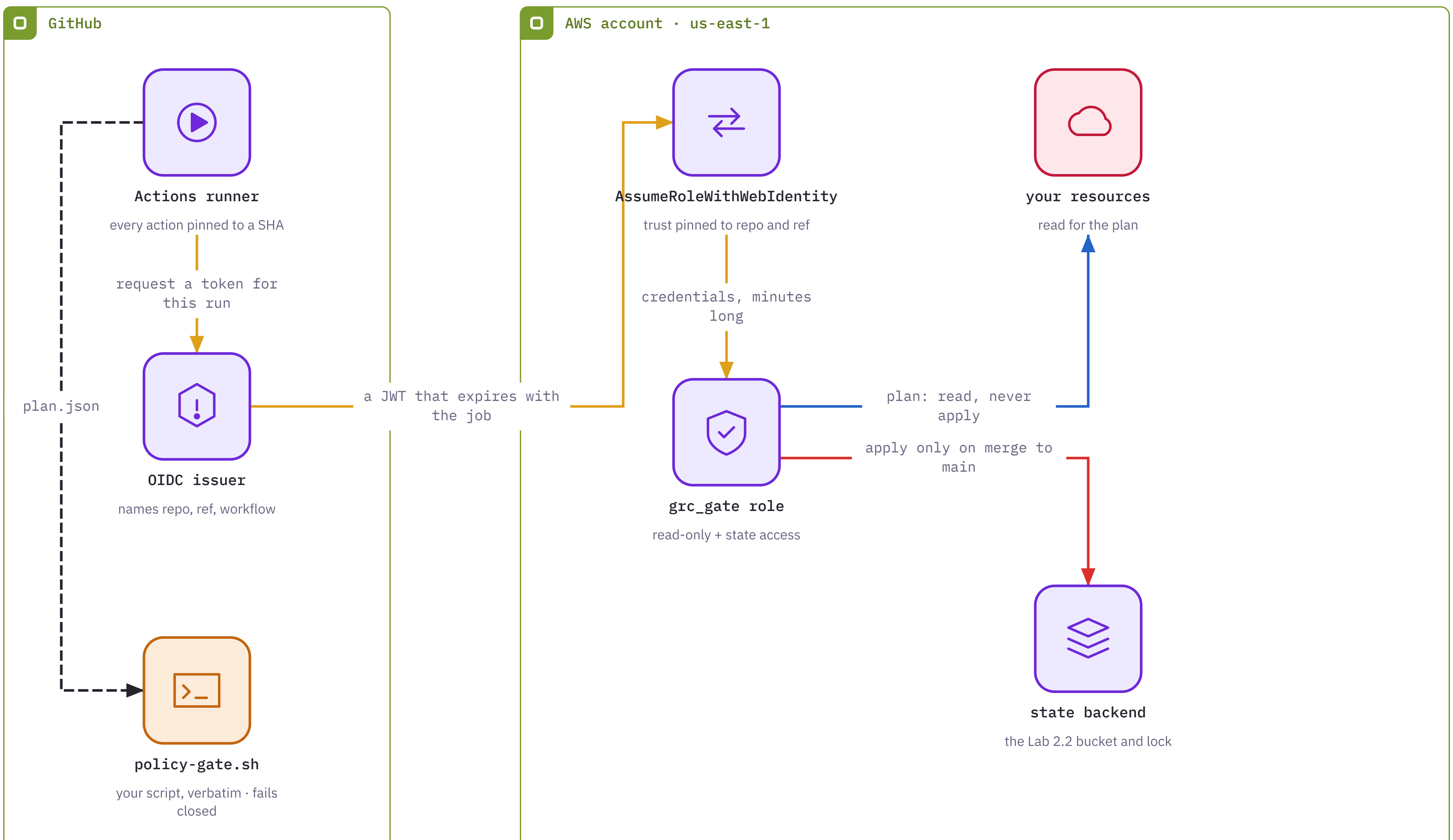
The same five controls against AWS plans, plus the script CI will call verbatim. Deliberately degraded copies of Lab 2.3 exist so you can watch the gate reject something.



Lab 4.3 — the gate on a machine you do not control

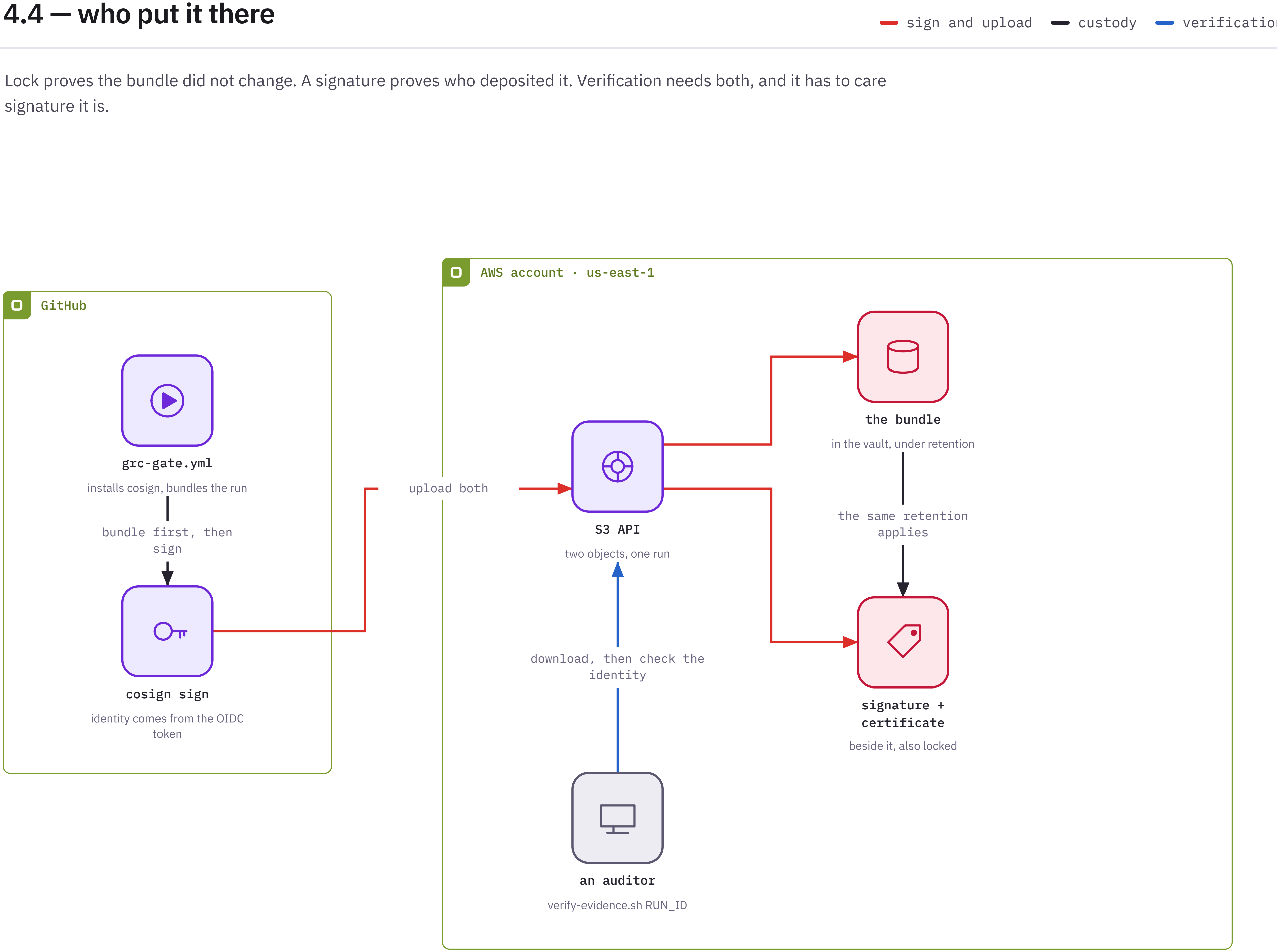
authentication · plan — read only · apply on merge · inside the runner

The runner has no stored credential. It asks GitHub for a token describing this exact run, trades that token for AWS credentials that last minutes, and gets a role scoped so it can read your infrastructure but not change it.



Lab 4.4 — who put it there

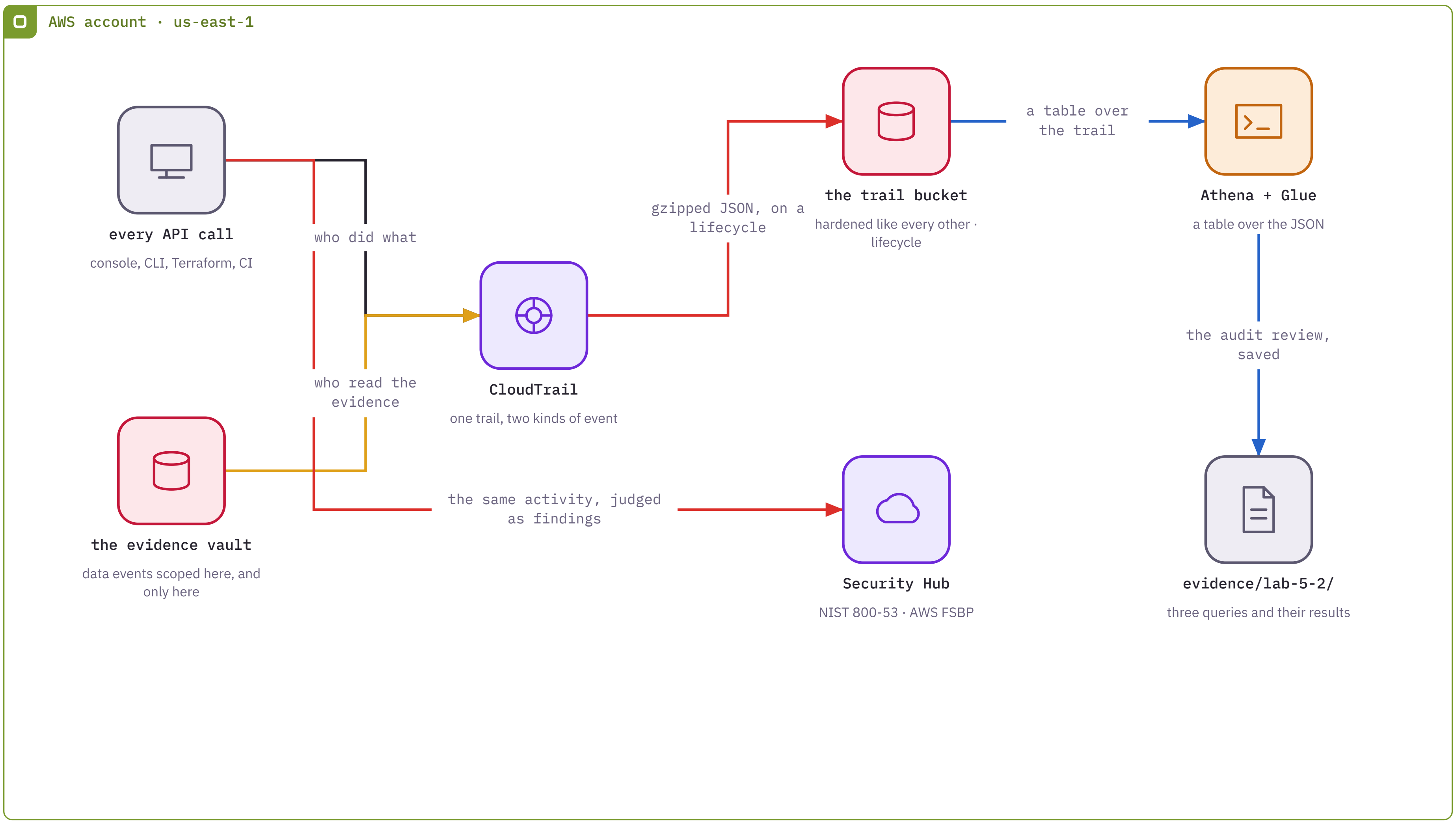
Object Lock proves the bundle did not change. A signature proves who deposited it. Verification needs both, and it has to care whose signature it is.



Lab 5.2 — the lab that earns AU-6

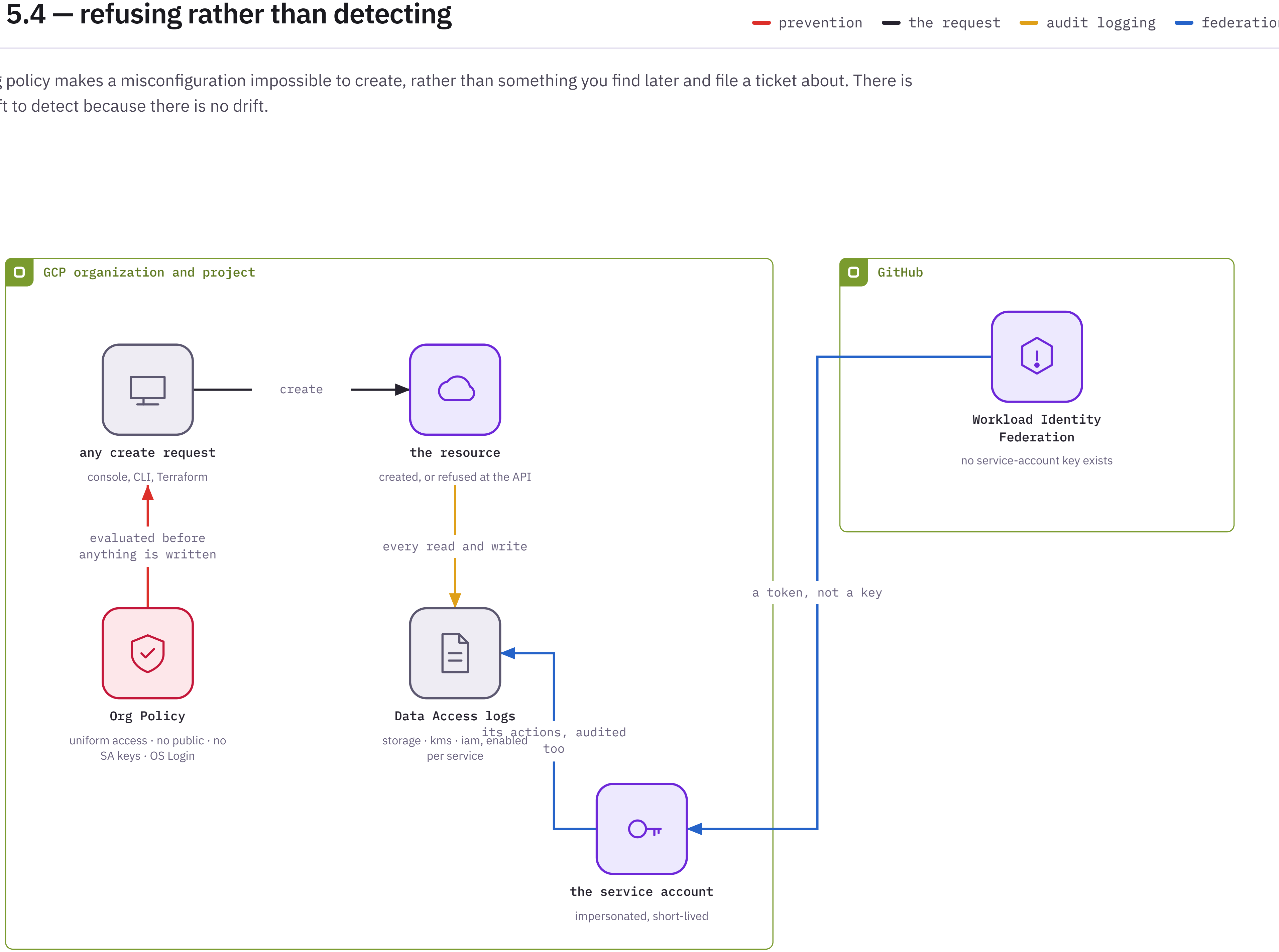
management events · data events · storage · the review

Until now logs were collected. Here something reads them. Shipping logs into a bucket is AU-3 plus AU-11; standing up a query over them is what AU-6 actually costs.



Lab 5.4 — refusing rather than detecting

An org policy makes a misconfiguration impossible to create, rather than something you find later and file a ticket about. There is no drift to detect because there is no drift.



Lab 6.1 — the document, and whether it is true

Everything the curriculum built becomes one document with links in it. trestle proves the document is well-formed. Only the graph verifier proves it describes anything real.